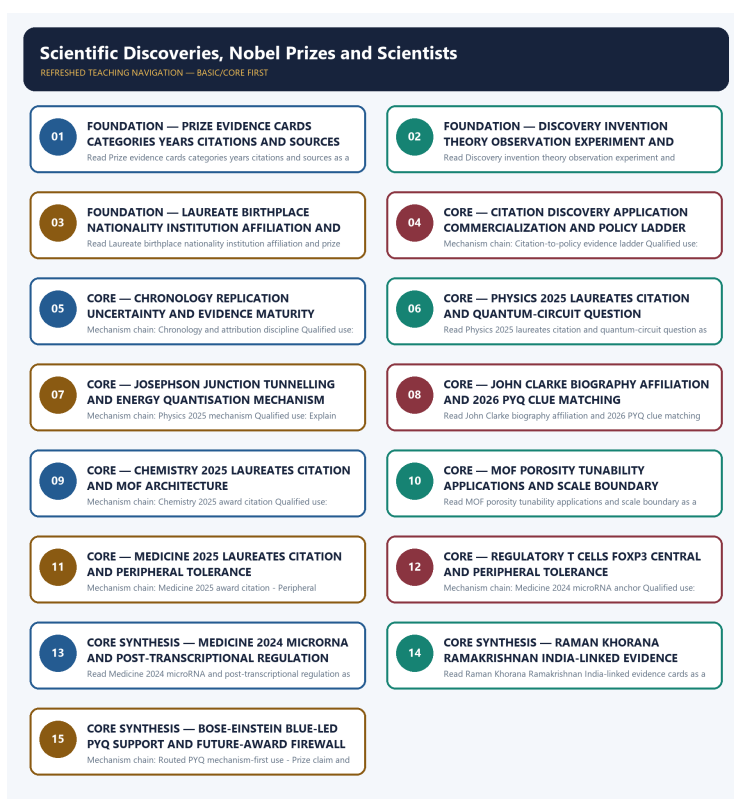


## COMPLETE LEARNING SESSION

# Scientific Discoveries, Nobel Prizes and Scientists - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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# Scientific Discoveries, Nobel Prizes and Scientists - Learner-v2 Complete Learning Session

Authoring-only generation: 2026-09-04. No PDF was rendered and no tracker or index was mutated.

## SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-04.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable official UPSC papers were supplementary only for printed routed demands. No answer key, performance value, procurement outcome, operational deployment, programme target or technology milestone was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** The audited ledgers route 2026 Prelims Q81 directly to this owner. The 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain directly owned by Topic 21 and appear only as cross-owner examples of attribution and citation-to-application discipline. Three representative cards preserve that boundary and reproduce no objective answer.
- **Live-link boundary:** Official Nobel Prize pages checked on 2026-09-04 confirmed completed 2025 Physics, Chemistry and Medicine awards, the 2024 microRNA award and the Raman, Khorana and Ramakrishnan evidence cards. Award citations, mechanisms, possible applications, trials, commercialization and policy outcomes remain explicitly separated; no 2026 laureate or prize outcome was invented before announcement.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

## LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-04. Substantive official text is used only for the proposition it supports; title-only, blocked or thin pages are recorded and supply no factual claim.

- <https://www.nobelprize.org/prizes/physics/2025/press-release/> and <https://www.nobelprize.org/prizes/physics/2025/clarke/facts/> - fetched 2026-09-04; official pages confirmed the laureates, 7 October 2025 citation, superconducting-circuit mechanism, John Clarke's birthplace and award-time affiliation. Opportunities for quantum technologies were not rewritten as a commercial computer, deployment or policy result.
- <https://www.nobelprize.org/prizes/chemistry/2025/press-release/> - fetched 2026-09-04; the official release confirmed Susumu Kitagawa, Richard Robson and Omar M. Yaghi and the citation for development of metal-organic frameworks. Listed possible uses were retained as material capabilities or applications, not proven commercialization.
- <https://www.nobelprize.org/prizes/medicine/2025/press-release/> - fetched 2026-09-04; the official release confirmed Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi and the citation concerning peripheral immune tolerance. Regulatory T cells, Foxp3 and clinical-trial references were not converted into approved therapies or outcomes.
- <https://www.nobelprize.org/prizes/medicine/2024/press-release/> - fetched 2026-09-04; the official release confirmed Victor Ambros and Gary Ruvkun and the citation for microRNA and post-transcriptional gene regulation. Discovery significance was kept separate from a diagnostic, medicine, commercial product or public-health outcome.
- <https://www.nobelprize.org/prizes/physics/1930/raman/facts/> - fetched 2026-09-04; the official page confirmed C. V. Raman's prize, Calcutta University affiliation, scattering-of-light citation and Raman-effect mechanism. Later analytical use was not inserted into the citation.
- <https://www.nobelprize.org/prizes/medicine/1968/khorana/facts/> and <https://www.nobelprize.org/prizes/chemistry/2009/ramakrishnan/facts/> - fetched 2026-09-04; official pages confirmed Har Gobind Khorana's genetic-code citation and Venkatraman Ramakrishnan's ribosome citation, with

their award-time affiliations. Birthplace, affiliation, field and national identity were not collapsed into one biographical label.

## BASIC LEARNING SESSION

### DEEP-REVIEW LEARNING CONTRACT

| Control             | Binding rule for this package                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Syllabus boundary   | Complete Science and Technology Basic/Core is easy-first and answer-complete before optional Advanced depth.                                              |
| System boundary     | Organisation, platform, subsystem, payload, process, application, institution and outcome remain distinct.                                                |
| Technical boundary  | Physical mechanism, classification, stage, platform, unit, range/capacity and limiting condition are explicit.                                            |
| Status boundary     | Announcement, approval, development, test, deployment, induction, commercial operation and observed outcome remain distinct.                              |
| Data boundary       | Every mission, launch, target, capacity, budget, range, timeline, rule and regulatory claim keeps source, date, unit and status.                          |
| Governance boundary | Funder, developer, operator, authoriser, regulator, procurer, settlement actor and adjudicator are not conflated.                                         |
| Practice contract   | Every solved item has demand decoding, detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement. |
| PYQ contract        | Official wording and keys are asserted only when verified; routed or reconstructed demands remain labelled.                                               |
| Dual-flow contract  | Graphical and ASCII masters independently reconstruct the same complete Basic-to-optional-Advanced evidence spine.                                        |
| Approval            | This immutable successor remains approved: false pending explicit approval.                                                                               |

**Canonical Basic/Core owner:** upsc-ai-kit\knowledge\Science-and-Technology\basic\26\_Scientific-Discoveries-Nobel-Prizes-and-Scientists.md **Substantive canonical provenance owner:** upsc-ai-kit\knowledge\Science-and-Technology\learning-sessions\v2\subject-wide-syllabus\science-and-technology-26\_Learning-Session.md **Optional Advanced owner:** upsc-ai-kit\knowledge\Science-and-Technology\advanced\26\_Scientific-Discoveries-Nobel-Prizes-and-Scientists.md  
**Official syllabus mapping:**  
 upsc-ai-kit\knowledge\Science-and-Technology\OFFICIAL-UPSC-SYLLABUS-MAPPING.md

### EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- ISRO organisation, launcher stages, payload and orbit claims retain their exact object and mission date.
- NavIC, GAGAN, human-spaceflight, planetary, nuclear, fusion, missile and procurement claims retain category and status.
- Aadhaar/UPI, AI, quantum and semiconductor claims retain actor, layer, metric, maturity and governance boundaries.
- DPDP/cybersecurity, biotechnology and genetic-engineering claims retain legal/regulatory stage and competent institution.
- Targets, budgets, capacities, ranges and timelines are never promoted into achieved outcomes without dated evidence.
- **Current-status note, rechecked 2026-09-06:** volatile claims retain source/date/status; failed official fetches remain explicit and supply no invented fact.

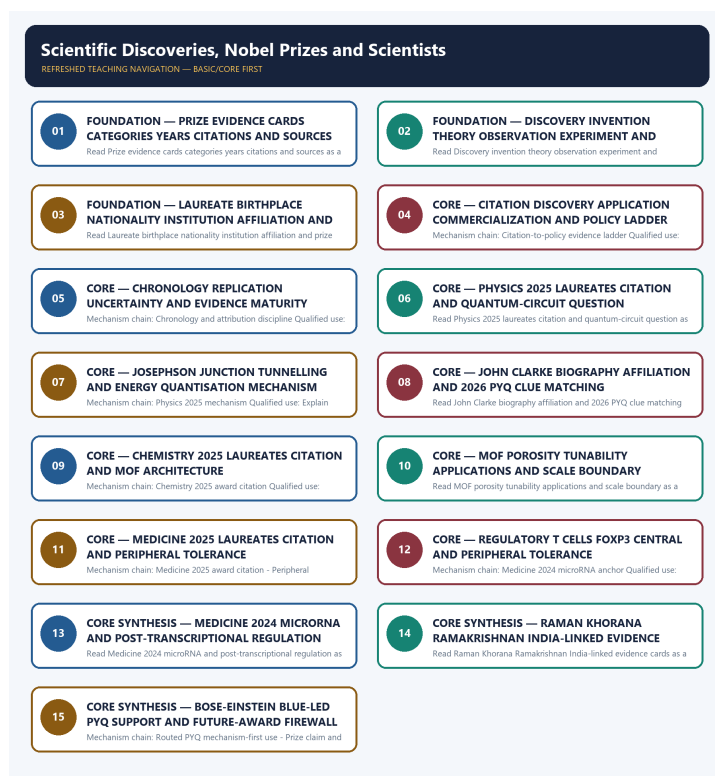
**Generation-local live/current sources:**

- <https://www.nobelprize.org/prizes/physics/2025/press-release/> and <https://www.nobelprize.org/prizes/physics/2025/clarke/facts/> - fetched 2026-09-04; official pages confirmed the laureates, 7 October 2025 citation, superconducting-circuit mechanism, John Clarke's birthplace and award-time affiliation. Opportunities for quantum technologies were not rewritten as a commercial computer, deployment or policy result.
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- <https://www.nobelprize.org/prizes/medicine/2025/press-release/> - fetched 2026-09-04; the official release confirmed Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi and the citation concerning peripheral immune tolerance. Regulatory T cells, Foxp3 and clinical-trial references were not converted into approved therapies or outcomes.
- <https://www.nobelprize.org/prizes/medicine/2024/press-release/> - fetched 2026-09-04; the official release confirmed Victor Ambros and Gary Ruvkun and the citation for microRNA and post-transcriptional gene regulation. Discovery significance was kept separate from a diagnostic, medicine, commercial product or public-health outcome.
- <https://www.nobelprize.org/prizes/physics/1930/raman/facts/> - fetched 2026-09-04; the official page confirmed C. V. Raman's prize, Calcutta University affiliation, scattering-of-light citation and Raman-effect mechanism. Later analytical use was not inserted into the citation.
- <https://www.nobelprize.org/prizes/medicine/1968/khorana/facts/> and <https://www.nobelprize.org/prizes/chemistry/2009/ramakrishnan/facts/> - fetched 2026-09-04; official pages confirmed Har Gobind Khorana's genetic-code citation and Venkatraman Ramakrishnan's ribosome citation, with their award-time affiliations. Birthplace, affiliation, field and national identity were not collapsed into one biographical label.

**Topic-routed verified PYQ owners:**

- No topic-local official question file is claimed; repository PYQ routing ledgers control wording and status.





Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

## SESSION 1 - FOUNDATION - PRIZE EVIDENCE CARDS CATEGORIES YEARS CITATIONS AND SOURCES

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not identify a laureate from a remembered field without matching category, year, citation and biography.

**Technical definition:** Read Prize evidence cards categories years citations and sources as a sequence: identify Minimum prize evidence card, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Prize evidence cards categories years citations and sources as a sequence: identify Minimum prize evidence card, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- FOUNDATION
- Prize evidence cards categories years citations
- sources
- Mechanism chain
- Qualified use
- Nobel

**How to use them:** Frame the answer through FOUNDATION; define Prize evidence cards categories years citations, connect sources with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

PRIZE EVIDENCE CARDS CATEGORIES YEARS CITATIONS AND SOURCES

01. Minimum prize evidence card

|

v

02. Discovery invention theory and observation

STATUS / CATEGORY FIREWALL -> Do not identify a laureate from a remembered field without matching category, year, citation and biography.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

## SIMPLE SYSTEM EXAMPLE

Read Prize evidence cards categories years citations and sources as a sequence: identify Minimum prize evidence card, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not identify a laureate from a remembered field without matching category, year, citation and biography. This keeps Minimum prize evidence card and Discovery invention theory and observation in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.
- A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

## EXAMINER CAUTION

- Do not identify a laureate from a remembered field without matching category, year, citation and biography.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Write the exact category, year, spelling, citation, mechanism and award-time affiliation before interpretation.

## MINI RECAP

- **Mechanism chain:** Minimum prize evidence card -> Discovery invention theory and observation
- **Qualified use:** Write the exact category, year, spelling, citation, mechanism and award-time affiliation before interpretation.

## CLOSING RECALL FLOW - FOUNDATION - PRIZE EVIDENCE CARDS CATEGORIES YEARS CITATIONS AND SOURCES

START / CONCEPT: FOUNDATION - Prize evidence cards categories years citations and sources

|  
v

EXACT TERMS: FOUNDATION • Prize evidence cards categories years citations • sources • Mechanism chain • Qualified use • Nobel

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Minimum prize evidence card - Discovery invention theory and observation Qualified use: Write the exact category, year, spelling, citation, mechanism and award-time affiliation before interpretation.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not identify a laureate from a remembered field without matching category, year, citation and biography.

|  
v

UPSC TRAP / ANSWER-USE: Do not identify a laureate from a remembered field without matching category, year, citation and biography.

|  
v

ANSWER-GRABBING FORMULATION: Read Prize evidence cards categories years citations and sources as a sequence: identify Minimum prize evidence card, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 2 - FOUNDATION - DISCOVERY INVENTION THEORY OBSERVATION EXPERIMENT AND ATTRIBUTION

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not use discovery, invention, theory, observation and application as synonyms.

**Technical definition:** Read Discovery invention theory observation experiment and attribution as a sequence: identify Nobel category architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Discovery invention theory observation experiment and attribution as a sequence: identify Nobel category architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- FOUNDATION
- Discovery invention theory observation experiment
- attribution
- Mechanism chain
- Qualified use
- The Nobel

**How to use them:** Frame the answer through FOUNDATION; define Discovery invention theory observation experiment, connect attribution with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

DISCOVERY INVENTION THEORY OBSERVATION EXPERIMENT AND ATTRIBUTION

01. Nobel category architecture

STATUS / CATEGORY FIREWALL -> Do not use discovery, invention, theory, observation and application as synonyms.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

## SIMPLE SYSTEM EXAMPLE

Read Discovery invention theory observation experiment and attribution as a sequence: identify Nobel category architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not use discovery, invention, theory, observation and application as synonyms. This keeps Nobel category architecture in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

## EXAMINER CAUTION

- Do not use discovery, invention, theory, observation and application as synonyms.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Identify whether the source recognises explanation, observation, discovery, method, development or invention.

## MINI RECAP

- **Mechanism chain:** Nobel category architecture
- **Qualified use:** Identify whether the source recognises explanation, observation, discovery, method, development or invention.

## CLOSING RECALL FLOW - FOUNDATION - DISCOVERY INVENTION THEORY OBSERVATION EXPERIMENT AND ATTRIBUTION

START / CONCEPT: FOUNDATION - Discovery invention theory observation experiment and attribution  
|  
v  
EXACT TERMS: FOUNDATION · Discovery invention theory observation experiment · attribution ·  
Mechanism chain · Qualified use · The Nobel  
|  
v  
MECHANISM / ARGUMENT: Mechanism chain: Nobel category architecture Qualified use: Identify whether  
the source recognises explanation, observation, discovery, method, development or invention.  
|  
v  
CONSEQUENCE / CONTRAST: The decisive boundary is Do not use discovery, invention, theory,  
observation and application as synonyms.  
|  
v  
UPSC TRAP / ANSWER-USE: Do not use discovery, invention, theory, observation and application as  
synonyms.  
|  
v  
ANSWER-GRABBING FORMULATION: Read Discovery invention theory observation experiment and attribution  
as a sequence: identify Nobel category architecture, follow the named mechanism to its user or  
output, and stop at the last verified status rather than assuming the next stage.

## SESSION 3 - FOUNDATION - LAUREATE BIRTHPLACE NATIONALITY INSTITUTION AFFILIATION AND PRIZE SHARING

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts.

**Technical definition:** Read Laureate birthplace nationality institution affiliation and prize sharing as a sequence: identify Laureate institution nationality boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Laureate birthplace nationality institution affiliation and prize sharing as a sequence: identify Laureate institution nationality boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- FOUNDATION
- Laureate birthplace nationality institution affiliation
- prize sharing
- Mechanism chain
- Qualified use
- Birthplace

**How to use them:** Frame the answer through FOUNDATION; define Laureate birthplace nationality institution affiliation, connect prize sharing with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

LAUREATE BIRTHPLACE NATIONALITY INSTITUTION AFFILIATION AND PRIZE SHARING

01. Laureate institution nationality boundary

STATUS / CATEGORY FIREWALL -> Do not call the Economic Sciences prize one of Alfred Nobel's original five.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

## SIMPLE SYSTEM EXAMPLE

Read Laureate birthplace nationality institution affiliation and prize sharing as a sequence: identify Laureate institution nationality boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not call the Economic Sciences prize one of Alfred Nobel's original five. This keeps Laureate institution nationality boundary in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

## EXAMINER CAUTION

- Do not call the Economic Sciences prize one of Alfred Nobel's original five.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Keep birthplace, citizenship, education, workplace, affiliation and prize share in separate fields.

## MINI RECAP

- **Mechanism chain:** Laureate institution nationality boundary
- **Qualified use:** Keep birthplace, citizenship, education, workplace, affiliation and prize share in separate fields.

## CLOSING RECALL FLOW - FOUNDATION - LAUREATE BIRTHPLACE NATIONALITY INSTITUTION AFFILIATION AND PRIZE SHARING

START / CONCEPT: FOUNDATION - Laureate birthplace nationality institution affiliation and prize sharing

|  
v

EXACT TERMS: FOUNDATION · Laureate birthplace nationality institution affiliation · prize sharing · Mechanism chain · Qualified use · Birthplace

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Laureate institution nationality boundary Qualified use: Keep birthplace, citizenship, education, workplace, affiliation and prize share in separate fields.

|  
v

CONSEQUENCE / CONTRAST: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not call the Economic Sciences prize one of Alfred Nobel's original five.

|  
v

ANSWER-GRABBING FORMULATION: Read Laureate birthplace nationality institution affiliation and prize sharing as a sequence: identify Laureate institution nationality boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 4 - CORE - CITATION DISCOVERY APPLICATION COMMERCIALIZATION AND POLICY LADDER

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Citation discovery application commercialization and policy ladder as a sequence: identify Citation-to-policy evidence ladder, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Citation-to-policy evidence ladder Qualified use: Move from citation to mechanism, application, engineering, commercialization and measured policy effect without skipping rungs.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect.

### MUST-WRITE KEYWORDS

- Citation discovery application commercialization
- policy ladder
- Mechanism chain
- Qualified use
- None
- Read Citation

**How to use them:** Frame the answer through Citation discovery application commercialization; define policy ladder, connect Mechanism chain with Qualified use to explain the mechanism, and use None for the decisive comparison or qualification.

### VISUAL FIRST

CITATION DISCOVERY APPLICATION COMMERCIALIZATION AND POLICY LADDER

01. Citation-to-policy evidence ladder

STATUS / CATEGORY FIREWALL -> Do not infer nationality from birthplace or award-time affiliation.

*The rail fixes the technical sequence and the claim boundary before explanation.*

### CORE EXPLANATION

The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

### SIMPLE SYSTEM EXAMPLE

Read Citation discovery application commercialization and policy ladder as a sequence: identify Citation-to-policy evidence ladder, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### TECHNICAL DISTINCTION

The decisive boundary is Do not infer nationality from birthplace or award-time affiliation. This keeps Citation-to-policy evidence ladder in the correct scientific category, institution and unit of measurement.

### NAMED EVIDENCE AND MECHANISM

- The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

### EXAMINER CAUTION

- Do not infer nationality from birthplace or award-time affiliation.

### EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Move from citation to mechanism, application, engineering, commercialization and measured policy effect without skipping rungs.

### MINI RECAP

- **Mechanism chain:** Citation-to-policy evidence ladder
- **Qualified use:** Move from citation to mechanism, application, engineering, commercialization and measured policy effect without skipping rungs.

### CLOSING RECALL FLOW - CORE - CITATION DISCOVERY APPLICATION COMMERCIALIZATION AND POLICY LADDER

START / CONCEPT: CORE - Citation discovery application commercialization and policy ladder

|  
v

EXACT TERMS: Citation discovery application commercialization · policy ladder · Mechanism chain · Qualified use · None · Read Citation

|  
v

MECHANISM / ARGUMENT: Read Citation discovery application commercialization and policy ladder as a sequence: identify Citation-to-policy evidence ladder, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|  
v

CONSEQUENCE / CONTRAST: Mechanism chain: Citation-to-policy evidence ladder Qualified use: Move from citation to mechanism, application, engineering, commercialization and measured policy effect without skipping rungs.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not infer nationality from birthplace or award-time affiliation.

|  
v

ANSWER-GRABBING FORMULATION: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect.

## SESSION 5 - CORE - CHRONOLOGY REPLICATION UNCERTAINTY AND EVIDENCE MATURITY

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Chronology replication uncertainty and evidence maturity as a sequence: identify Chronology and attribution discipline, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Chronology and attribution discipline Qualified use: Date theory, experiment, replication, award and later technology separately and qualify uncertainty.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Chronology replication uncertainty and evidence maturity as a sequence: identify Chronology and attribution discipline, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- Chronology replication uncertainty



- **evidence maturity**
- **Mechanism chain**
- **Qualified use**
- **Prize year**
- **Prize**

**How to use them:** Frame the answer through Chronology replication uncertainty; define evidence maturity, connect Mechanism chain with Qualified use to explain the mechanism, and use Prize year for the decisive comparison or qualification.

## VISUAL FIRST

CHRONOLOGY REPLICATION UNCERTAINTY AND EVIDENCE MATURITY

01. Chronology and attribution discipline

STATUS / CATEGORY FIREWALL -> Do not rewrite a later application into the official prize citation.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

## SIMPLE SYSTEM EXAMPLE

Read Chronology replication uncertainty and evidence maturity as a sequence: identify Chronology and attribution discipline, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not rewrite a later application into the official prize citation. This keeps Chronology and attribution discipline in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

## EXAMINER CAUTION

- Do not rewrite a later application into the official prize citation.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Date theory, experiment, replication, award and later technology separately and qualify uncertainty.

## MINI RECAP

- **Mechanism chain:** Chronology and attribution discipline
- **Qualified use:** Date theory, experiment, replication, award and later technology separately and qualify uncertainty.

## CLOSING RECALL FLOW - CORE - CHRONOLOGY REPLICATION UNCERTAINTY AND EVIDENCE MATURITY

START / CONCEPT: CORE - Chronology replication uncertainty and evidence maturity

|  
v

EXACT TERMS: Chronology replication uncertainty · evidence maturity · Mechanism chain · Qualified use · Prize year · Prize

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Chronology and attribution discipline Qualified use: Date theory, experiment, replication, award and later technology separately and qualify uncertainty.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not rewrite a later application into the official prize citation.

|  
v

UPSC TRAP / ANSWER-USE: Do not rewrite a later application into the official prize citation.

|  
v

ANSWER-GRABBING FORMULATION: Read Chronology replication uncertainty and evidence maturity as a sequence: identify Chronology and attribution discipline, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 6 - CORE - PHYSICS 2025 LAUREATES CITATION AND QUANTUM-CIRCUIT QUESTION

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Technical definition:** Read Physics 2025 laureates citation and quantum-circuit question as a sequence: identify Replication uncertainty and mature evidence, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Physics 2025 laureates citation and quantum-circuit question as a sequence: identify Replication uncertainty and mature evidence, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- Physics 2025 laureates citation
- quantum-circuit question
- Mechanism chain
- Qualified use
- Strong
- Nobel Prize

**How to use them:** Frame the answer through Physics 2025 laureates citation; define quantum-circuit question, connect Mechanism chain with Qualified use to explain the mechanism, and use Strong for the decisive comparison or qualification.

## VISUAL FIRST

PHYSICS 2025 LAUREATES CITATION AND QUANTUM-CIRCUIT QUESTION

01. Replication uncertainty and mature evidence

|  
v

02. Physics 2025 award citation

STATUS / CATEGORY FIREWALL -> Do not treat prize year as automatically the discovery, publication, patent or deployment year.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

## SIMPLE SYSTEM EXAMPLE

Read Physics 2025 laureates citation and quantum-circuit question as a sequence: identify Replication uncertainty and mature evidence, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not treat prize year as automatically the discovery, publication, patent or deployment year. This keeps Replication uncertainty and mature evidence and Physics 2025 award citation in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.
- The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

## EXAMINER CAUTION

- Do not treat prize year as automatically the discovery, publication, patent or deployment year.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Preserve the Physics citation and all three laureates before using the quantum-technology link.

## MINI RECAP

- **Mechanism chain:** Replication uncertainty and mature evidence -> Physics 2025 award citation
- **Qualified use:** Preserve the Physics citation and all three laureates before using the quantum-technology link.

## CLOSING RECALL FLOW - CORE - PHYSICS 2025 LAUREATES CITATION AND QUANTUM-CIRCUIT QUESTION

START / CONCEPT: CORE - Physics 2025 laureates citation and quantum-circuit question

|  
v

EXACT TERMS: Physics 2025 laureates citation · quantum-circuit question · Mechanism chain · Qualified use · Strong · Nobel Prize

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Replication uncertainty and mature evidence - Physics 2025 award citation Qualified use: Preserve the Physics citation and all three laureates before using the quantum-technology link.

|  
v

CONSEQUENCE / CONTRAST: Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

|  
v

UPSC TRAP / ANSWER-USE: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints.

|  
v

ANSWER-GRABBING FORMULATION: Read Physics 2025 laureates citation and quantum-circuit question as a sequence: identify Replication uncertainty and mature evidence, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 7 - CORE - JOSEPHSON JUNCTION TUNNELLING AND ENERGY QUANTISATION MECHANISM

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Josephson junction tunnelling and energy quantisation mechanism as a sequence: identify Physics 2025 mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Physics 2025 mechanism Qualified use: Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Mechanism chain: Physics 2025 mechanism Qualified use: Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

### MUST-WRITE KEYWORDS

- Josephson junction tunnelling
- energy quantisation mechanism
- Mechanism chain
- Qualified use
- Read Josephson
- Physics

**How to use them:** Frame the answer through Josephson junction tunnelling; define energy quantisation mechanism, connect Mechanism chain with Qualified use to explain the mechanism, and use Read Josephson for the decisive comparison or qualification.

## VISUAL FIRST

JOSEPHSON JUNCTION TUNNELLING AND ENERGY QUANTISATION MECHANISM

01. Physics 2025 mechanism

STATUS / CATEGORY FIREWALL -> Do not claim a Nobel Prize proves replication, scale, affordability or policy success.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

## SIMPLE SYSTEM EXAMPLE

Read Josephson junction tunnelling and energy quantisation mechanism as a sequence: identify Physics 2025 mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not claim a Nobel Prize proves replication, scale, affordability or policy success. This keeps Physics 2025 mechanism in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

## EXAMINER CAUTION

- Do not claim a Nobel Prize proves replication, scale, affordability or policy success.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

## MINI RECAP

- **Mechanism chain:** Physics 2025 mechanism
- **Qualified use:** Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

## CLOSING RECALL FLOW - CORE - JOSEPHSON JUNCTION TUNNELLING AND ENERGY QUANTISATION MECHANISM

START / CONCEPT: CORE - Josephson junction tunnelling and energy quantisation mechanism

|  
v

EXACT TERMS: Josephson junction tunnelling · energy quantisation mechanism · Mechanism chain · Qualified use · Read Josephson · Physics

|  
v

MECHANISM / ARGUMENT: Read Josephson junction tunnelling and energy quantisation mechanism as a sequence: identify Physics 2025 mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|  
v

CONSEQUENCE / CONTRAST: Mains: Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not claim a Nobel Prize proves replication, scale, affordability or policy success.

|  
v

ANSWER-GRABBING FORMULATION: Mechanism chain: Physics 2025 mechanism Qualified use: Explain superconductivity, Josephson junction, collective state, tunnelling and quantised energy in sequence.

## SESSION 8 - CORE - JOHN CLARKE BIOGRAPHY AFFILIATION AND 2026 PYQ CLUE MATCHING

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Technical definition:** Read John Clarke biography affiliation and 2026 PYQ clue matching as a sequence: identify John Clarke biographical route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read John Clarke biography affiliation and 2026 PYQ clue matching as a sequence: identify John Clarke biographical route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- John Clarke biography affiliation
- Mechanism chain
- Qualified use
- Birthplace and United States institutional affiliation
- Birthplace
- 2026 PYQ clue matching

**How to use them:** Frame the answer through John Clarke biography affiliation; define Mechanism chain, connect Qualified use with Birthplace and United States institutional affiliation to explain the mechanism, and use Birthplace for the decisive comparison or qualification.

## VISUAL FIRST

JOHN CLARKE BIOGRAPHY AFFILIATION AND 2026 PYQ CLUE MATCHING

01. John Clarke biographical route

STATUS / CATEGORY FIREWALL -> Do not reduce the 2025 Physics citation to the generic phrase quantum computing.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

## SIMPLE SYSTEM EXAMPLE

Read John Clarke biography affiliation and 2026 PYQ clue matching as a sequence: identify John Clarke biographical route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not reduce the 2025 Physics citation to the generic phrase quantum computing. This keeps John Clarke biographical route in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

## EXAMINER CAUTION

- Do not reduce the 2025 Physics citation to the generic phrase quantum computing.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Match Cambridge birthplace and Berkeley affiliation without treating either as nationality proof.

## MINI RECAP

- **Mechanism chain:** John Clarke biographical route
- **Qualified use:** Match Cambridge birthplace and Berkeley affiliation without treating either as nationality proof.

## CLOSING RECALL FLOW - CORE - JOHN CLARKE BIOGRAPHY AFFILIATION AND 2026 PYQ CLUE MATCHING

START / CONCEPT: CORE - John Clarke biography affiliation and 2026 PYQ clue matching

|

v

EXACT TERMS: John Clarke biography affiliation · Mechanism chain · Qualified use · Birthplace and United States institutional affiliation · Birthplace · 2026 PYQ clue matching

|

v

MECHANISM / ARGUMENT: Mechanism chain: John Clarke biographical route Qualified use: Match Cambridge birthplace and Berkeley affiliation without treating either as nationality proof.

|

v

CONSEQUENCE / CONTRAST: Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not reduce the 2025 Physics citation to the generic phrase quantum computing.

|

v

ANSWER-GRABBING FORMULATION: Read John Clarke biography affiliation and 2026 PYQ clue matching as a sequence: identify John Clarke biographical route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

# SESSION 9 - CORE - CHEMISTRY 2025 LAUREATES CITATION AND MOF ARCHITECTURE

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Chemistry 2025 laureates citation and MOF architecture as a sequence: identify Chemistry 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Chemistry 2025 award citation Qualified use: Preserve the Chemistry citation and distinguish material architecture from one use case.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Chemistry 2025 laureates citation and MOF architecture as a sequence: identify Chemistry 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## MUST-WRITE KEYWORDS

- Chemistry 2025 laureates citation
- MOF architecture
- Mechanism chain
- Qualified use
- Nobel Prize
- Chemistry

**How to use them:** Frame the answer through Chemistry 2025 laureates citation; define MOF architecture, connect Mechanism chain with Qualified use to explain the mechanism, and use Nobel Prize for the decisive comparison or qualification.

## VISUAL FIRST

CHEMISTRY 2025 LAUREATES CITATION AND MOF ARCHITECTURE

01. Chemistry 2025 award citation

STATUS / CATEGORY FIREWALL -> Do not call the superconducting-circuit experiment a deployed fault-tolerant quantum computer.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

## SIMPLE SYSTEM EXAMPLE

Read Chemistry 2025 laureates citation and MOF architecture as a sequence: identify Chemistry 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not call the superconducting-circuit experiment a deployed fault-tolerant quantum computer. This keeps Chemistry 2025 award citation in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.



### EXAMINER CAUTION

- Do not call the superconducting-circuit experiment a deployed fault-tolerant quantum computer.

### EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Preserve the Chemistry citation and distinguish material architecture from one use case.

### MINI RECAP

- **Mechanism chain:** Chemistry 2025 award citation
- **Qualified use:** Preserve the Chemistry citation and distinguish material architecture from one use case.

### CLOSING RECALL FLOW - CORE - CHEMISTRY 2025 LAUREATES CITATION AND MOF ARCHITECTURE

START / CONCEPT: CORE - Chemistry 2025 laureates citation and MOF architecture

|

v

EXACT TERMS: Chemistry 2025 laureates citation · MOF architecture · Mechanism chain · Qualified use  
· Nobel Prize · Chemistry

|

v

MECHANISM / ARGUMENT: Mechanism chain: Chemistry 2025 award citation Qualified use: Preserve the Chemistry citation and distinguish material architecture from one use case.

|

v

CONSEQUENCE / CONTRAST: Mains: Preserve the Chemistry citation and distinguish material architecture from one use case.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not call the superconducting-circuit experiment a deployed fault-tolerant quantum computer.

|

v

ANSWER-GRABBING FORMULATION: Read Chemistry 2025 laureates citation and MOF architecture as a sequence: identify Chemistry 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 10 - CORE - MOF POROSITY TUNABILITY APPLICATIONS AND SCALE BOUNDARY

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not infer sole authorship from one laureate's biography in a shared prize.

**Technical definition:** Read MOF porosity tunability applications and scale boundary as a sequence: identify Metal-organic framework mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read MOF porosity tunability applications and scale boundary as a sequence: identify Metal-organic framework mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- MOF porosity tunability applications
- scale boundary
- Mechanism chain
- Qualified use
- Metal-organic
- Official

**How to use them:** Frame the answer through MOF porosity tunability applications; define scale boundary, connect Mechanism chain with Qualified use to explain the mechanism, and use Metal-organic for the decisive comparison or qualification.

## VISUAL FIRST

MOF POROSITY TUNABILITY APPLICATIONS AND SCALE BOUNDARY

01. Metal-organic framework mechanism

STATUS / CATEGORY FIREWALL -> Do not infer sole authorship from one laureate's biography in a shared prize.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

## SIMPLE SYSTEM EXAMPLE

Read MOF porosity tunability applications and scale boundary as a sequence: identify Metal-organic framework mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not infer sole authorship from one laureate's biography in a shared prize. This keeps Metal-organic framework mechanism in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

## EXAMINER CAUTION

- Do not infer sole authorship from one laureate's biography in a shared prize.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Connect metal nodes, organic linkers, pores and tunability, then stop before unsupported scale claims.

## MINI RECAP

- **Mechanism chain:** Metal-organic framework mechanism
- **Qualified use:** Connect metal nodes, organic linkers, pores and tunability, then stop before unsupported scale claims.

## CLOSING RECALL FLOW - CORE - MOF POROSITY TUNABILITY APPLICATIONS AND SCALE BOUNDARY

START / CONCEPT: CORE - MOF porosity tunability applications and scale boundary

|

v

EXACT TERMS: MOF porosity tunability applications · scale boundary · Mechanism chain · Qualified use · Metal-organic · Official

|

v

MECHANISM / ARGUMENT: Mechanism chain: Metal-organic framework mechanism Qualified use: Connect metal nodes, organic linkers, pores and tunability, then stop before unsupported scale claims.

|

v

CONSEQUENCE / CONTRAST: Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not infer sole authorship from one laureate's biography in a shared prize.

|

v

ANSWER-GRABBING FORMULATION: Read MOF porosity tunability applications and scale boundary as a sequence: identify Metal-organic framework mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 11 - CORE - MEDICINE 2025 LAUREATES CITATION AND PERIPHERAL TOLERANCE

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Medicine 2025 laureates citation and peripheral tolerance as a sequence: identify Medicine 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Medicine 2025 award citation - Peripheral tolerance mechanism Qualified use: Preserve the Medicine citation and identify the self-tolerance problem before naming applications.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Medicine 2025 laureates citation and peripheral tolerance as a sequence: identify Medicine 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **Medicine 2025 laureates citation**
- **peripheral tolerance**
- **Mechanism chain**
- **Qualified use**
- **Nobel Prize**
- **Physiology**

**How to use them:** Frame the answer through Medicine 2025 laureates citation; define peripheral tolerance, connect Mechanism chain with Qualified use to explain the mechanism, and use Nobel Prize for the decisive comparison or qualification.

## VISUAL FIRST

MEDICINE 2025 LAUREATES CITATION AND PERIPHERAL TOLERANCE

01. Medicine 2025 award citation

|  
v

02. Peripheral tolerance mechanism

STATUS / CATEGORY FIREWALL -> Do not convert metal-organic-framework capability into commercialization or environmental outcome.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

## SIMPLE SYSTEM EXAMPLE

Read Medicine 2025 laureates citation and peripheral tolerance as a sequence: identify Medicine 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not convert metal-organic-framework capability into commercialization or environmental outcome. This keeps Medicine 2025 award citation and Peripheral tolerance mechanism in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
- Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

## EXAMINER CAUTION

- Do not convert metal-organic-framework capability into commercialization or environmental outcome.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Preserve the Medicine citation and identify the self-tolerance problem before naming applications.

## MINI RECAP

- **Mechanism chain:** Medicine 2025 award citation -> Peripheral tolerance mechanism
- **Qualified use:** Preserve the Medicine citation and identify the self-tolerance problem before naming applications.

## CLOSING RECALL FLOW - CORE - MEDICINE 2025 LAUREATES CITATION AND PERIPHERAL TOLERANCE

START / CONCEPT: CORE - Medicine 2025 laureates citation and peripheral tolerance

|  
v

EXACT TERMS: Medicine 2025 laureates citation · peripheral tolerance · Mechanism chain · Qualified use · Nobel Prize · Physiology

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Medicine 2025 award citation - Peripheral tolerance mechanism Qualified use: Preserve the Medicine citation and identify the self-tolerance problem before naming applications.

|  
v

CONSEQUENCE / CONTRAST: These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not convert metal-organic-framework capability into commercialization or environmental outcome.

|  
v

ANSWER-GRABBING FORMULATION: Read Medicine 2025 laureates citation and peripheral tolerance as a sequence: identify Medicine 2025 award citation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 12 - CORE - REGULATORY T CELLS FOXP3 CENTRAL AND PERIPHERAL TOLERANCE

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.

**Technical definition:** Mechanism chain: Medicine 2024 microRNA anchor Qualified use: Trace regulatory T cells and Foxp3 while separating central from peripheral tolerance and trial from approval.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Mechanism chain: Medicine 2024 microRNA anchor Qualified use: Trace regulatory T cells and Foxp3 while separating central from peripheral tolerance and trial from approval.

### MUST-WRITE KEYWORDS

- Regulatory T cells Foxp3 central
- peripheral tolerance
- Mechanism chain
- Qualified use
- RNAs
- RNA

**How to use them:** Frame the answer through Regulatory T cells Foxp3 central; define peripheral tolerance, connect Mechanism chain with Qualified use to explain the mechanism, and use RNAs for the decisive comparison or qualification.

## VISUAL FIRST

REGULATORY T CELLS FOXP3 CENTRAL AND PERIPHERAL TOLERANCE

01. Medicine 2024 microRNA anchor

STATUS / CATEGORY FIREWALL -> Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

## SIMPLE SYSTEM EXAMPLE

Read Regulatory T cells Foxp3 central and peripheral tolerance as a sequence: identify Medicine 2024 microRNA anchor, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim. This keeps Medicine 2024 microRNA anchor in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

## EXAMINER CAUTION

- Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Trace regulatory T cells and Foxp3 while separating central from peripheral tolerance and trial from approval.

## MINI RECAP

- **Mechanism chain:** Medicine 2024 microRNA anchor
- **Qualified use:** Trace regulatory T cells and Foxp3 while separating central from peripheral tolerance and trial from approval.

## CLOSING RECALL FLOW - CORE - REGULATORY T CELLS FOXP3 CENTRAL AND PERIPHERAL TOLERANCE

START / CONCEPT: CORE - Regulatory T cells Foxp3 central and peripheral tolerance

|  
v

EXACT TERMS: Regulatory T cells Foxp3 central · peripheral tolerance · Mechanism chain · Qualified use · RNAs · RNA

|  
v

MECHANISM / ARGUMENT: Read Regulatory T cells Foxp3 central and peripheral tolerance as a sequence: identify Medicine 2024 microRNA anchor, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.

|  
v

UPSC TRAP / ANSWER-USE: Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.

|  
v

ANSWER-GRABBING FORMULATION: Mechanism chain: Medicine 2024 microRNA anchor Qualified use: Trace regulatory T cells and Foxp3 while separating central from peripheral tolerance and trial from approval.

## SESSION 13 - CORE SYNTHESIS - MEDICINE 2024 MICRORNA AND POST-TRANSCRIPTIONAL REGULATION

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not turn microRNA association into a diagnostic or medicine without evidence.

**Technical definition:** Read Medicine 2024 microRNA and post-transcriptional regulation as a sequence: identify Raman prize and mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Medicine 2024 microRNA and post-transcriptional regulation as a sequence: identify Raman prize and mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Medicine 2024 microRNA
- post-transcriptional regulation
- Mechanism chain
- Qualified use
- The Raman

**How to use them:** Frame the answer through CORE SYNTHESIS; define Medicine 2024 microRNA, connect post-transcriptional regulation with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

MEDICINE 2024 MICRORNA AND POST-TRANSCRIPTIONAL REGULATION

01. Raman prize and mechanism

STATUS / CATEGORY FIREWALL -> Do not turn microRNA association into a diagnostic or medicine without evidence.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

## SIMPLE SYSTEM EXAMPLE

Read Medicine 2024 microRNA and post-transcriptional regulation as a sequence: identify Raman prize and mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not turn microRNA association into a diagnostic or medicine without evidence. This keeps Raman prize and mechanism in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

## EXAMINER CAUTION

- Do not turn microRNA association into a diagnostic or medicine without evidence.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Explain microRNA action after transcription and before protein output without therapeutic overreach.

## MINI RECAP

- **Mechanism chain:** Raman prize and mechanism
- **Qualified use:** Explain microRNA action after transcription and before protein output without therapeutic overreach.



## CLOSING RECALL FLOW - CORE SYNTHESIS - MEDICINE 2024 MICRORNA AND POST-TRANSCRIPTIONAL REGULATION

START / CONCEPT: CORE SYNTHESIS - Medicine 2024 microRNA and post-transcriptional regulation

|  
v

EXACT TERMS: CORE SYNTHESIS · Medicine 2024 microRNA · post-transcriptional regulation · Mechanism chain · Qualified use · The Raman

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Raman prize and mechanism Qualified use: Explain microRNA action after transcription and before protein output without therapeutic overreach.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not turn microRNA association into a diagnostic or medicine without evidence.

|  
v

UPSC TRAP / ANSWER-USE: Do not turn microRNA association into a diagnostic or medicine without evidence.

|  
v

ANSWER-GRABBING FORMULATION: Read Medicine 2024 microRNA and post-transcriptional regulation as a sequence: identify Raman prize and mechanism, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 14 - CORE SYNTHESIS - RAMAN KHORANA RAMAKRISHNAN INDIA-LINKED EVIDENCE CARDS

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.

**Technical definition:** Read Raman Khorana Ramakrishnan India-linked evidence cards as a sequence: identify Khorana genetic-code prize, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Raman Khorana Ramakrishnan India-linked evidence cards as a sequence: identify Khorana genetic-code prize, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Raman Khorana Ramakrishnan India-linked evidence cards
- Mechanism chain
- Qualified use
- X-ray crystallography mapped ribosomal structure; antibiotic relevance
- X-ray

**How to use them:** Frame the answer through CORE SYNTHESIS; define Raman Khorana Ramakrishnan India-linked evidence cards, connect Mechanism chain with Qualified use to explain the mechanism, and use X-ray crystallography mapped ribosomal structure; antibiotic relevance for the decisive comparison or qualification.

## VISUAL FIRST

RAMAN KHORANA RAMAKRISHNAN INDIA-LINKED EVIDENCE CARDS

01. Khorana genetic-code prize

|  
v

02. Ramakrishnan ribosome prize

STATUS / CATEGORY FIREWALL -> Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

## SIMPLE SYSTEM EXAMPLE

Read Raman Khorana Ramakrishnan India-linked evidence cards as a sequence: identify Khorana genetic-code prize, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution. This keeps Khorana genetic-code prize and Ramakrishnan ribosome prize in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.
- Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

## EXAMINER CAUTION

- Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Build separate prize cards for Raman, Khorana and Ramakrishnan with category, affiliation and mechanism.

## MINI RECAP

- **Mechanism chain:** Khorana genetic-code prize -> Ramakrishnan ribosome prize
- **Qualified use:** Build separate prize cards for Raman, Khorana and Ramakrishnan with category, affiliation and mechanism.

## CLOSING RECALL FLOW - CORE SYNTHESIS - RAMAN KHORANA RAMAKRISHNAN INDIA-LINKED EVIDENCE CARDS

START / CONCEPT: CORE SYNTHESIS - Raman Khorana Ramakrishnan India-linked evidence cards  
|  
v  
EXACT TERMS: CORE SYNTHESIS · Raman Khorana Ramakrishnan India-linked evidence cards · Mechanism chain · Qualified use · X-ray crystallography mapped ribosomal structure; antibiotic relevance · X-ray  
|  
v  
MECHANISM / ARGUMENT: Mechanism chain: Khorana genetic-code prize - Ramakrishnan ribosome prize  
Qualified use: Build separate prize cards for Raman, Khorana and Ramakrishnan with category, affiliation and mechanism.  
|  
v  
CONSEQUENCE / CONTRAST: The decisive boundary is Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.  
|  
v  
UPSC TRAP / ANSWER-USE: Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.  
|  
v  
ANSWER-GRABBING FORMULATION: Read Raman Khorana Ramakrishnan India-linked evidence cards as a sequence: identify Khorana genetic-code prize, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## SESSION 15 - CORE SYNTHESIS - BOSE-EINSTEIN BLUE-LED PYQ SUPPORT AND FUTURE-AWARD FIREWALL

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Bose-Einstein blue-LED PYQ support and future-award firewall as a sequence: identify Routed PYQ mechanism-first use, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Routed PYQ mechanism-first use - Prize claim and current-status firewall  
Qualified use: Keep direct and cross-owner PYQ routes distinct and stop before unannounced 2026 prizes.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Bose-Einstein blue-LED PYQ support and future-award firewall as a sequence: identify Routed PYQ mechanism-first use, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Bose-Einstein blue-LED PYQ support
- future-award firewall
- Mechanism chain
- Qualified use
- Subject

**How to use them:** Frame the answer through CORE SYNTHESIS; define Bose-Einstein blue-LED PYQ support, connect future-award firewall with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

BOSE-EINSTEIN BLUE-LED PYQ SUPPORT AND FUTURE-AWARD FIREWALL

01. Routed PYQ mechanism-first use

|

v

02. Prize claim and current-status firewall

STATUS / CATEGORY FIREWALL -> Do not invent 2026 laureates before official announcements.

*The rail fixes the technical sequence and the claim boundary before explanation.*

## CORE EXPLANATION

The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

## SIMPLE SYSTEM EXAMPLE

Read Bose-Einstein blue-LED PYQ support and future-award firewall as a sequence: identify Routed PYQ mechanism-first use, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not invent 2026 laureates before official announcements. This keeps Routed PYQ mechanism-first use and Prize claim and current-status firewall in the correct scientific category, institution and unit of measurement.

## NAMED EVIDENCE AND MECHANISM

- The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.
- Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

## EXAMINER CAUTION

- Do not invent 2026 laureates before official announcements.

## EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Keep direct and cross-owner PYQ routes distinct and stop before unannounced 2026 prizes.

## MINI RECAP

- **Mechanism chain:** Routed PYQ mechanism-first use -> Prize claim and current-status firewall
- **Qualified use:** Keep direct and cross-owner PYQ routes distinct and stop before unannounced 2026 prizes.

## COMPLETE BASIC OWNER EVIDENCE BANK

**Subject:** Science & Technology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III + Prelims. **Core area:** Identifying scientists through discovery, discipline, nationality, institutional affiliation and prize-year clues without reducing science current affairs to personality trivia. **Grounded in:** NobelPrize.org prize-category and laureate pages; the 2025 Nobel Prize in Physics facts page for John Clarke; NCERT science foundations; official 2026 Prelims Set-A scan. Current facts verified 2 Aug 2026. **FACT:** = source-grounded | **ANALYSIS:** = analytical inference | **CURRENT:** = dated current anchor | **WRONG:** = trap. *Companion:* advanced/26\_Scientific-Discoveries-Nobel-Prizes-and-Scientists.md.

### 1. Visual identification method

BIOGRAPHICAL CLUE  
(birthplace / nationality)  
+  
INSTITUTIONAL CLUE  
(university / laboratory at award)  
+  
SCIENTIFIC CLUE  
(field / discovery / technology)  
+  
PRIZE-YEAR CLUE  
|  
v  
IDENTIFY THE LAUREATE -> LINK BACK TO THE UNDERLYING SCIENCE

**Core proposition:** A laureate question is not answered safely by memorising names alone. Match every name to the prize category, award year, discovery and affiliation stated by the official prize source.

### 2. Nobel architecture

| Category                     | Core subject link                                                      | Frequent trap                                                                   |
|------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| FACT: Physics                | Matter, energy, quantum science, astronomy, electronics                | A technological application may rest on a much older physical discovery.        |
| FACT: Chemistry              | Molecules, materials, catalysis, biochemistry                          | Chemistry prizes often overlap biology but remain Chemistry awards.             |
| FACT: Physiology or Medicine | Biological mechanism, disease and therapy                              | Institution or nationality does not identify the scientific contribution.       |
| FACT: Literature             | Literary body of work                                                  | Not a science category.                                                         |
| FACT: Peace                  | Peace, rights, humanitarian or institutional contribution              | Can be awarded to an organisation as well as a person.                          |
| FACT: Economic Sciences      | Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel | It was established later and is not one of Alfred Nobel's original five prizes. |

**FACT:** The awarding institutions differ by category. Do not infer a category merely from the word "Nobel" or from the laureate's employer.

### 3. Minimum data card for every current laureate

| Field                  | What to record                                                        |
|------------------------|-----------------------------------------------------------------------|
| Prize                  | Exact category and year                                               |
| Laureate               | Official spelling                                                     |
| Motivation             | Discovery or contribution in the official citation                    |
| Science mechanism      | Two-line explanation in ordinary language                             |
| Birth clue             | Place/country only when officially stated                             |
| Award-time affiliation | Institution at the time of award, not every past employer             |
| India link             | Record only if direct and source-supported                            |
| Status date            | Verification date because affiliations and biographies can be updated |

### 4. 2025 Physics case tested in 2026

CURRENT: **John Clarke - Nobel Prize in Physics 2025**

- FACT: Born in **Cambridge, United Kingdom**, in 1942.
- FACT: Affiliated at the time of the award with the **University of California, Berkeley, USA**.
- FACT: Shared the prize with **Michel H. Devoret** and **John M. Martinis**.
- FACT: Official motivation: discovery of **macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit**.
- FACT: Their superconducting electrical circuit demonstrated quantum behaviour at a macroscopic scale.
- ANALYSIS: Exam link: connect the biography to superconductivity, Josephson-junction circuits, quantum tunnelling and quantum-computing hardware; do not stop at the name.

### 5. Discipline-to-topic routing

| Discovery clue                                 | Specialist owner to revise                          |
|------------------------------------------------|-----------------------------------------------------|
| Quantum circuit, tunnelling, superconductivity | Topic 21 physics + Topic 10 quantum technology      |
| Protein structure, molecular design, catalysis | Topic 22 chemistry and relevant biotechnology topic |
| Gene regulation, immunity, disease mechanism   | Topic 23 biology + Topics 13-15                     |
| AI or neural-network foundations               | Topic 09 AI plus the originating science discipline |
| Climate, materials or energy discovery         | Topics 18, 20-22 as applicable                      |

### 6. Must-know elimination rules

- FACT: **Birthplace, citizenship and award-time affiliation are different facts**.
- FACT: A laureate may work in a country other than the country of birth.
- FACT: A prize can be shared; one option may name a co-laureate whose biography does not match the question.
- FACT: Match the **year** before matching a remembered discovery.
- FACT: Match the official **motivation**, not a broad popular label.
- WRONG: A university affiliation proves nationality. -> It proves only the stated institutional association.
- WRONG: Every famous scientist connected with a field won that year's Nobel. -> Use year plus citation plus biography.
- WRONG: A current-affairs name list is sufficient. -> The underlying mechanism remains examinable after the name becomes stale.

## Core answer architecture - discovery attribution and mechanism-first current affairs

**Thesis choice.** A discovery/prize question is an evidence-matching task: official citation, year, laureate, award-time affiliation and mechanism must align; popularity, nationality or a company announcement cannot fill a gap.

**10-mark spine.** Start with the exact official prize/discovery fact; explain the underlying mechanism in two lines; connect it to the appropriate Physics/Chemistry/Biology/technology owner; state the status boundary between finding and application.

**15/20-mark spine.** Use **problem and method -> named discoverer/team and evidence -> mechanism -> downstream applications and Indian relevance -> uncertainty, replication, maturity and ethics**. Do not turn a scientist biography into a technology policy answer.

### Evidence units.

- **Claim:** prize attribution is narrower than field attribution -> **official Nobel category, year and motivation** -> identifies what was recognised rather than a broad public label -> **qualification:** affiliation at award is not nationality, and an employer does not prove authorship of every prior contribution.
- **Claim:** a scientific discovery matters through a mechanism -> **the 2025 Physics citation on macroscopic quantum tunnelling and energy quantisation in an electric circuit** -> routes a biography clue to superconducting circuits, tunnelling and quantum hardware -> **qualification:** recognition of fundamental science does not establish a commercial quantum computer or national deployment.
- **Claim:** current discovery evidence has a maturity ladder -> **finding -> replication/engineering -> regulated application -> scale** -> prevents a company claim, preprint or demonstration from being written as settled societal capability -> **qualification:** do not invent discovery dates, prize details, efficacy or future uses absent the official source.

**Verdict.** Use current names as entry points to durable scientific mechanisms and retain source/year/status discipline in every attribution.

## CLOSING RECALL FLOW - CORE SYNTHESIS - BOSE-EINSTEIN BLUE-LED PYQ SUPPORT AND FUTURE-AWARD FIREWALL

START / CONCEPT: CORE SYNTHESIS - Bose-Einstein blue-LED PYQ support and future-award firewall

|  
v

EXACT TERMS: CORE SYNTHESIS · Bose-Einstein blue-LED PYQ support · future-award firewall ·  
Mechanism chain · Qualified use · Subject

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Routed PYQ mechanism-first use - Prize claim and current-status firewall Qualified use: Keep direct and cross-owner PYQ routes distinct and stop before unannounced 2026 prizes.

|  
v

CONSEQUENCE / CONTRAST: Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

|  
v

UPSC TRAP / ANSWER-USE: The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

|  
v

ANSWER-GRABBING FORMULATION: Read Bose-Einstein blue-LED PYQ support and future-award firewall as a sequence: identify Routed PYQ mechanism-first use, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

## TRANSITION EVIDENCE AND PERIODIZATION LEDGER - TOPIC 26

### Models and criteria

| Model                                  | Core use                                                            | Limit                                                          |
|----------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------|
| Colonial Hindu/Muslim/British division | history of periodization                                            | communal ruler-religion label cannot explain social transition |
| Dynastic chronology                    | orders Gupta, post-Gupta and regional polities                      | ruler change is not simultaneous structural change             |
| Indian feudalism                       | grants, intermediaries, immunities, labour and exchange contraction | outcomes and commerce vary by region                           |
| Segmentary state                       | graded ritual/political control beyond a core                       | derived mainly from later-south debates; not universal         |
| Integrative/regional state             | chiefs, grants, cults and local elites build regional power         | integration remains unequal and coercive                       |

Test transition through polity, agrarian relations, revenue/labour, settlement, exchange, institutions, social classification, religion, language and material culture. The c. 300-550, 550-750 and 750-1000 phases are heuristics, not universal ruptures.

### Grant rights and ground effects

| Recipient/right                                   | Historical question                                     | Source caution                                |
|---------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|
| Brahmana, temple, monastery, official or retainer | why was revenue or land assigned?                       | recipient classes vary by region and date     |
| taxes and produce                                 | who collected which dues?                               | exemption formula is not proof of enforcement |
| labour (vishti)                                   | who supplied transport, construction or provisioning?   | incidence must be locally evidenced           |
| judicial/fiscal immunity                          | were fines, officials or policing functions restricted? | specified right is not total sovereignty      |
| water, pasture, trees and forest                  | which older users were affected?                        | charter silence does not mean empty land      |

## REGIONAL CONTINUITY-CHANGE LEDGER - TOPIC 26

| Region                | Transition pattern                                                             | Anti-universalization rule                     |
|-----------------------|--------------------------------------------------------------------------------|------------------------------------------------|
| North                 | post-Gupta courts, samantas, grants and selected urban contraction             | no instantaneous anarchy                       |
| East                  | agrarian expansion, Pala-Sena formations and mahavihara networks               | no uniform Buddhist society                    |
| Western India         | regional lineages, ports and Jain/Brahmanical institutions                     | genealogy is not literal ethnicity             |
| Deccan                | Vakataka-Chalukya-Rashtrakuta grants, temples, irrigation and exchange         | no one feudal sequence                         |
| South                 | Pallava/Pandya and Chola-boundary brahmadeyas, tanks, temples and local bodies | full Imperial Chola system belongs to Topic 27 |
| Forest/hill frontiers | chiefs, products, cultivation, cults, resistance and peasantization            | incorporation is not passive assimilation      |

### Social, institutional and knowledge controls

- Samanta changed from neighbouring ruler to varied subordinate/intermediary usage; it was not one legal class.



- Temples and monasteries could be landholders, employers, redistributors, archives, schools and political nodes, but functions varied locally.
- Varna prescription, jati process, occupation, untouchability and exclusion must be separated by source, region and date.
- Gender analysis must include property, marriage, labour, donor/political and religious agency without one all-India status verdict.
- Bhakti was not uniformly egalitarian; Buddhism and Jainism transformed regionally rather than simply disappearing.
- Sanskrit cosmopolitan continuity coexisted with vernacular literary and epigraphic growth.
- Incoming peoples affected political and cultural formation, but invasion alone did not cause the transition.

### PYQ ownership control

The repository routes **zero direct PYQs** to Topic 26. The ten retained questions belong to Topics 10, 20, 22, 23, 27 or Indian Art and Culture and are adjacent continuity/change practice only.

**Final method:** model -> evidence class -> regional case -> continuity -> transformation -> source limit -> qualified verdict.

## SCIENCE AND TECHNOLOGY DEEP-REVIEW CORE CONTROL

- **Must remember:** Scientific-history questions separate observation, discovery, theory, experiment, invention and application; Nobel claims require category, award year, laureate, official motivation and the exact nature of any Indian connection.
- **Close distinction:** Discovery is not invention, theory is not proof beyond revision, award year is not publication year, nomination is not award, institution is not nationality, Indian origin is not automatically an award to India, and prediction is not result.
- **Mechanism / status / evidence limit:** Use Nobel Foundation or equally authoritative dated evidence for current awards; state contribution and later application without anachronism, and distinguish announcement, experimental confirmation, recognition and observed impact.

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly identifies Minimum prize evidence card?

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: A. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner,**

## Q2. Which option preserves the technical boundary of Minimum prize evidence card?

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: B. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Q3. Which statement uses Minimum prize evidence card without changing its institution, unit or status?

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: C. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Q4. Which option avoids the standard UPSC close-option trap about Minimum prize evidence card?

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect

was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: D. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q5. Which statement correctly identifies Discovery invention theory and observation?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: A. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q6. Which option preserves the technical boundary of Discovery invention theory and observation?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: B. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q7. Which statement uses Discovery invention theory and observation without changing its institution, unit or status?

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: C. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q8. Which option avoids the standard UPSC close-option trap about Discovery invention theory and observation?

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: D. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q9. Which statement correctly identifies Nobel category architecture?

A. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or

author of every contribution. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: A. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q10. Which option preserves the technical boundary of Nobel category architecture?

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: B. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q11. Which statement uses Nobel category architecture without changing its institution, unit or status?

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: C. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q12. Which option avoids the standard UPSC close-option trap about Nobel category architecture?**

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: D. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q13. Which statement correctly identifies Laureate institution nationality boundary?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: A. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q14. Which option preserves the technical boundary of Laureate institution nationality boundary?**

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: B. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q15. Which statement uses Laureate institution nationality boundary without changing its institution, unit or status?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: C. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q16. Which option avoids the standard UPSC close-option trap about Laureate institution nationality boundary?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole

authorship nor nationality. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: D. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q17. Which statement correctly identifies Citation-to-policy evidence ladder?

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: A. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q18. Which option preserves the technical boundary of Citation-to-policy evidence ladder?

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q19. Which statement uses Citation-to-policy evidence ladder without changing its institution, unit or status?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: C. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q20. Which option avoids the standard UPSC close-option trap about Citation-to-policy evidence ladder?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: D. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q21. Which statement correctly identifies Chronology and attribution discipline?**

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. A single result, preprint, company announcement or prize does not eliminate

measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: A. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q22. Which option preserves the technical boundary of Chronology and attribution discipline?**

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q23. Which statement uses Chronology and attribution discipline without changing its institution, unit or status?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: C. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Q24. Which option avoids the standard UPSC close-option trap about Chronology and attribution discipline?

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: D. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Q25. Which statement correctly identifies Replication uncertainty and mature evidence?

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: A. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## Q26. Which option preserves the technical boundary of Replication uncertainty and mature evidence?

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum

tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q27. Which statement uses Replication uncertainty and mature evidence without changing its institution, unit or status?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: C. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q28. Which option avoids the standard UPSC close-option trap about Replication uncertainty and mature evidence?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: D. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q29. Which statement correctly identifies Physics 2025 award citation?

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: A. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q30. Which option preserves the technical boundary of Physics 2025 award citation?

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: B. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q31. Which statement uses Physics 2025 award citation without changing its institution, unit or status?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.



**Answer: C. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q32. Which option avoids the standard UPSC close-option trap about Physics 2025 award citation?**

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: D. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q33. Which statement correctly identifies Physics 2025 mechanism?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: A. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q34. Which option preserves the technical boundary of Physics 2025 mechanism?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q35. Which statement uses Physics 2025 mechanism without changing its institution, unit or status?

A. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: C. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q36. Which option avoids the standard UPSC close-option trap about Physics 2025 mechanism?

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum

tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: D. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q37. Which statement correctly identifies John Clarke biographical route?

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: A. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q38. Which option preserves the technical boundary of John Clarke biographical route?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q39. Which statement uses John Clarke biographical route without changing its institution, unit or status?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: C. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q40. Which option avoids the standard UPSC close-option trap about John Clarke biographical route?**

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: D. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q41. Which statement correctly identifies Chemistry 2025 award citation?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. Shimon Sakaguchi identified

regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: A. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q42. Which option preserves the technical boundary of Chemistry 2025 award citation?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q43. Which statement uses Chemistry 2025 award citation without changing its institution, unit or status?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: C. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q44. Which option avoids the standard UPSC close-option trap about Chemistry 2025 award citation?

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: D. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q45. Which statement correctly identifies Metal-organic framework mechanism?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: A. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### Q46. Which option preserves the technical boundary of Metal-organic framework mechanism?

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not

prove approved treatment or population outcome. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: B. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q47. Which statement uses Metal-organic framework mechanism without changing its institution, unit or status?**

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: C. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q48. Which option avoids the standard UPSC close-option trap about Metal-organic framework mechanism?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: D. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport,**

catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

**Q49. Which statement correctly identifies Medicine 2025 award citation?**

A. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: A. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q50. Which option preserves the technical boundary of Medicine 2025 award citation?**

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: B. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q51. Which statement uses Medicine 2025 award citation without changing its institution, unit or status?**

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q52. Which option avoids the standard UPSC close-option trap about Medicine 2025 award citation?**

A. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: D. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q53. Which statement correctly identifies Peripheral tolerance mechanism?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a

therapy. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: A. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q54. Which option preserves the technical boundary of Peripheral tolerance mechanism?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

**Answer: B. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q55. Which statement uses Peripheral tolerance mechanism without changing its institution, unit or status?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q56. Which option avoids the standard UPSC close-option trap about Peripheral tolerance mechanism?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: D. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q57. Which statement correctly identifies Medicine 2024 microRNA anchor?**

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: A. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q58. Which option preserves the technical boundary of Medicine 2024 microRNA anchor?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function

in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: B. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q59. Which statement uses Medicine 2024 microRNA anchor without changing its institution, unit or status?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q60. Which option avoids the standard UPSC close-option trap about Medicine 2024 microRNA anchor?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: D. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q61. Which statement correctly identifies Raman prize and mechanism?

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: A. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q62. Which option preserves the technical boundary of Raman prize and mechanism?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: B. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q63. Which statement uses Raman prize and mechanism without changing its institution, unit or status?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The direct 2026 routed demand identifies a 2025 laureate

from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: C. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q64. Which option avoids the standard UPSC close-option trap about Raman prize and mechanism?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. D. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

**Answer: D. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q65. Which statement correctly identifies Khorana genetic-code prize?**

A. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: A. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



### Q66. Which option preserves the technical boundary of Khorana genetic-code prize?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: B. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q67. Which statement uses Khorana genetic-code prize without changing its institution, unit or status?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: C. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q68. Which option avoids the standard UPSC close-option trap about Khorana genetic-code prize?

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded

verb and object must be preserved exactly. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: D. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q69. Which statement correctly identifies Ramakrishnan ribosome prize?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: A. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q70. Which option preserves the technical boundary of Ramakrishnan ribosome prize?**

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: B. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q71. Which statement uses Ramakrishnan ribosome prize without changing its institution, unit or status?**

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: C. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q72. Which option avoids the standard UPSC close-option trap about Ramakrishnan ribosome prize?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: D. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q73. Which statement correctly identifies Routed PYQ mechanism-first use?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather



than inventing future laureates. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: A. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q74. Which option preserves the technical boundary of Routed PYQ mechanism-first use?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: B. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q75. Which statement uses Routed PYQ mechanism-first use without changing its institution, unit or status?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: C. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q76. Which option avoids the standard UPSC close-option trap about Routed PYQ mechanism-first use?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: D. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q77. Which statement correctly identifies Prize claim and current-status firewall?**

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: A. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q78. Which option preserves the technical boundary of Prize claim and current-status firewall?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or

nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: B. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q79. Which statement uses Prize claim and current-status firewall without changing its institution, unit or status?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: C. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q80. Which option avoids the standard UPSC close-option trap about Prize claim and current-status firewall?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: D. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

## PYQS AND ANSWER PRACTICE

### AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The audited ledgers route 2026 Prelims Q81 directly to this owner. The 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain directly owned by Topic 21 and appear only as cross-owner examples of attribution and citation-to-application discipline. Three representative cards preserve that boundary and reproduce no objective answer.

### PYQ DEMAND CARD 1 - 2026 Prelims GS-I

**Demand:** Identify the 2025 Nobel Prize recipient by matching biographical and professional clues to the official category, citation and award-time affiliation.

**Status:** Verified direct routed objective demand covering 2026 Q81; the locally held Set-A key is provisional, so no option, answer letter or inferred key is supplied.

**Model solution: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** Treat "PYQ DEMAND CARD 1 - 2026 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

#### Detailed examiner-grade model answer:

**Introduction and thesis: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### Analytical body:

1. **Claim and named evidence:** Demand: Identify the 2025 Nobel Prize recipient by matching biographical and professional clues to the official category, citation and award-time affiliation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence.  
**Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified direct routed objective demand covering 2026 Q81; the locally held Set-A key is provisional, so no option, answer letter or inferred key is supplied. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence.  
**Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

**Why this earns marks:** It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

**How to improve this answer:** For "PYQ DEMAND CARD 1 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

### PYQ DEMAND CARD 2 - 2018 GS-III

**Demand:** Use citation and discovery-attribution discipline while discussing the contribution of Bose-Einstein statistics to physics.

**Status:** Cross-owner supporting card: Topic 21 retains the substantive statistics and physics answer; Topic 26 contributes attribution, chronology and discovery-to-application boundaries without claiming a Nobel citation for the routed concept.

**Model solution: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application



shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2018 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

#### **Analytical body:**

1. **Claim and named evidence:** Demand: Use citation and discovery-attribution discipline while discussing the contribution of Bose-Einstein statistics to physics. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.



**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 2 - 2018 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

### PYQ DEMAND CARD 3 - 2021 GS-III

**Demand:** Explain the everyday-life impact of the blue LED invention recognised by the 2014 Nobel Prize while separating award citation, mechanism, application and outcome.

**Status:** Cross-owner supporting card: Topic 21 retains semiconductor and LED physics; Topic 26 supplies prize-citation, invention, application and commercialization discipline rather than an official model answer.

**Model solution: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on "PYQ DEMAND CARD 3 - 2021 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each

later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

#### Analytical body:

1. **Claim and named evidence:** Demand: Explain the everyday-life impact of the blue LED invention recognised by the 2014 Nobel Prize while separating award citation, mechanism, application and outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Cross-owner supporting card: Topic 21 retains semiconductor and LED physics; Topic 26 supplies prize-citation, invention, application and commercialization discipline rather than an official model answer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 3 - 2021 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 1 - 10 MARKS

**Question:** Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical clues.

Answer in about 150 words.

**Model thesis: Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.
- The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.
- Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Qualified conclusion: Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion

retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **explain** requires a direct position on "Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** **Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Analytical body:**

1. **Claim and named evidence:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language



mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 2 - 10 MARKS

**Question:** Distinguish theory, observation, experiment, discovery, invention and development in prize attribution. Answer in about 150 words.

**Model thesis:** **Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.
- Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.
- A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Qualified conclusion: Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **answer** requires a direct position on "Distinguish theory, observation, experiment, discovery, invention and development in prize...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population



or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Distinguish theory, observation, experiment, discovery, invention and development in prize...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

### ORIGINAL MAINS 3 - 15 MARKS

**Question:** Analyse the 2025 Physics prize through its official citation, superconducting-circuit mechanism and technology boundary. Answer in about 250 words.

**Model thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.
- The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.
- The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution,

application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **analyse** requires a direct position on "Analyse the 2025 Physics prize through its official citation, superconducting-circuit...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis:** **Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### **Analytical body:**

1. **Claim and named evidence:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** Connect mechanism -> named system/institution -> application or implementation

pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Analyse the 2025 Physics prize through its official citation, superconducting-circuit...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from possible applications and commercialization. Answer in about 250 words.

**Model thesis: Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit,



institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.
- Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.
- The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
- Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Qualified conclusion:** **Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific

process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **discuss** requires a direct position on "Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### Detailed examiner-grade model answer:

**Introduction and thesis:** **Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Peripheral tolerance mechanism.

**Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or



strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism.

**Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune tolerance and microRNA reveal the path from basic science to application. Answer in about 300 words.

**Model thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

- The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.
- The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.
- Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.
- The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
- Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.
- The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy

implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **examine** requires a direct position on "Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis:** **Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune



vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

7. **Claim and named evidence:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution,



system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Evaluate discovery attribution and Indian scientific relevance through Raman, Khorana, Ramakrishnan and routed PYQs. Answer in about 300 words.

**Model thesis: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of

scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

- Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.
- Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.
- The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.
- Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Qualified conclusion: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize.

**Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use.

**Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **evaluate** requires a direct position on "Evaluate discovery attribution and Indian scientific relevance through Raman, Khorana,...", each clause, the exact technical mechanism, named Indian

institution, dated status, application, risk and qualification.

#### Detailed examiner-grade model answer:

**Introduction and thesis:** **Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped

ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and

development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Evaluate discovery attribution and Indian scientific relevance through Raman, Khorana,...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

**Subject:** Science & Technology | **Tier:** Advanced | **GS Paper:** GS-III + Prelims. **Purpose:** Convert prize current affairs into deeper scientific reasoning without making optional enrichment necessary for objective-question coverage. *Companion:* basic/26\_Scientific-Discoveries-Nobel-Prizes-and-Scientists.md.

### 1. Analytical lens

**ANALYSIS:** Major prizes reveal how science moves from a counter-intuitive mechanism to an experimental demonstration and then to a technology platform. The useful sequence is:

theory -> measurable phenomenon -> reproducible experiment  
-> enabling instrument/device -> engineering platform -> social application

For the 2025 Physics case, quantum tunnelling and energy quantisation were not merely abstract ideas: superconducting circuits made macroscopic quantum behaviour measurable and helped establish a route toward superconducting qubits and precision sensors.

### 2. Deeper distinctions

| Distinction                                   | Why it matters                                                                                                                             |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Discovery vs invention                        | A discovery establishes a phenomenon; an invention creates a device or process. Prize citations may recognise either kind of contribution. |
| Foundational science vs commercial technology | Scientific proof does not establish scalability, affordability or deployment.                                                              |
| Laureate vs institution                       | The person receives the prize; the institution supplies the research environment but is not automatically a co-recipient.                  |
| Citation vs later application                 | The official citation names the awarded work; later applications should be labelled as consequences, not rewritten into the citation.      |
| Individual prize vs organisational prize      | Peace prizes may recognise organisations; science prizes are awarded to individuals under the applicable rules.                            |

### 3. Mains-ready use

- **ANALYSIS:** Use laureate examples to illustrate long-horizon public research, instrumentation, international mobility and the path from basic science to strategic technology.
- **ANALYSIS:** Avoid "Nobel count" nationalism. A stronger answer examines research funding, universities, laboratory autonomy, interdisciplinary teams and translation capacity.
- **ANALYSIS:** Connect Topic 24's funding architecture with the specialist science file that explains the discovery.

### 4. Advanced trap

**WRONG: A prize proves that the resulting technology is mature.** **FACT:** It proves recognition of the cited contribution; engineering readiness, manufacturing scale, cost and regulation remain separate tests.



## CONSOLIDATED REGISTER NOTES

### Scientific Discoveries, Nobel Prizes and Scientists: PRIZE CATEGORY, LAUREATE, CITATION AND BIOGRAPHY EVIDENCE CARDS

1. **Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.
2. **Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.
3. **Nobel category architecture:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.
4. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.
5. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.
6. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.
7. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.
8. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.
9. **Physics 2025 mechanism:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.
10. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.
11. **Chemistry 2025 award citation:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.
12. **Metal-organic framework mechanism:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.



13. **Medicine 2025 award citation:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
14. **Peripheral tolerance mechanism:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.
15. **Medicine 2024 microRNA anchor:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.
16. **Raman prize and mechanism:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.
17. **Khorana genetic-code prize:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.
18. **Ramakrishnan ribosome prize:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.
19. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.
20. **Prize claim and current-status firewall:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

## Scientific Discoveries, Nobel Prizes and Scientists: 2025 PHYSICS, CHEMISTRY AND MEDICINE MECHANISM MAP

- Do not identify a laureate from a remembered field without matching category, year, citation and biography.
- Do not use discovery, invention, theory, observation and application as synonyms.
- Do not call the Economic Sciences prize one of Alfred Nobel's original five.
- Do not infer nationality from birthplace or award-time affiliation.
- Do not rewrite a later application into the official prize citation.
- Do not treat prize year as automatically the discovery, publication, patent or deployment year.
- Do not claim a Nobel Prize proves replication, scale, affordability or policy success.
- Do not reduce the 2025 Physics citation to the generic phrase quantum computing.
- Do not call the superconducting-circuit experiment a deployed fault-tolerant quantum computer.
- Do not infer sole authorship from one laureate's biography in a shared prize.
- Do not convert metal-organic-framework capability into commercialization or environmental outcome.
- Do not turn peripheral immune tolerance or clinical trials into an approved therapy claim.
- Do not turn microRNA association into a diagnostic or medicine without evidence.
- Do not collapse Raman, Khorana or Ramakrishnan birthplace, affiliation, category and contribution.

- Do not invent 2026 laureates before official announcements.

## Scientific Discoveries, Nobel Prizes and Scientists: RAMAN, KHORANA, RAMAKRISHNAN AND MICRO-RNA RAPID RECALL

WRITE CATEGORY YEAR OFFICIAL SPELLING CITATION AND AWARD-TIME AFFILIATION  
 -> SEPARATE BIRTHPLACE NATIONALITY EDUCATION WORKPLACE AND PRIZE SHARE  
 -> CLASSIFY THEORY OBSERVATION EXPERIMENT DISCOVERY INVENTION OR DEVELOPMENT  
 -> DATE DISCOVERY PUBLICATION REPLICATION AWARD PATENT PRODUCT AND DEPLOYMENT SEPARATELY  
 -> TRACE CITATION MECHANISM APPLICATION ENGINEERING COMMERCIALIZATION AND POLICY OUTCOME  
 -> EXPLAIN PHYSICS 2025 THROUGH SUPERCONDUCTING CIRCUIT JOSEPHSON JUNCTION AND TUNNELLING  
 -> EXPLAIN CHEMISTRY 2025 THROUGH METAL NODES ORGANIC LINKERS PORES AND TUNABILITY  
 -> EXPLAIN MEDICINE 2025 THROUGH REGULATORY T CELLS Foxp3 AND PERIPHERAL TOLERANCE  
 -> LINK MICRO-RNA TO POST-TRANSCRIPTIONAL REGULATION WITHOUT THERAPY OVERREACH  
 -> BUILD DISTINCT RAMAN KHORANA AND RAMAKRISHNAN PRIZE CARDS  
 -> KEEP BOSE-EINSTEIN AND BLUE-LED SUBSTANTIVE SCIENCE WITH TOPIC 21  
 -> STOP AT THE LATEST OFFICIALLY ANNOUNCED PRIZE AND LAST VERIFIED MATURITY RUNG

## Scientific Discoveries, Nobel Prizes and Scientists: PYQ ATTRIBUTION, APPLICATION-MATURITY AND FUTURE-AWARD FIREWALL

Official Nobel Prize pages checked on 2026-09-04 confirmed completed 2025 Physics, Chemistry and Medicine awards, the 2024 microRNA award and the Raman, Khorana and Ramakrishnan evidence cards. Award citations, mechanisms, possible applications, trials, commercialization and policy outcomes remain explicitly separated; no 2026 laureate or prize outcome was invented before announcement.

## COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

### ASCII MASTER FLOW - PANEL 1/12: Minimum prize evidence card

CATEGORY + YEAR  
 OFFICIAL LAUREATE SPELLING  
 OFFICIAL CITATION / MOTIVATION  
 MECHANISM + AWARD-TIME AFFILIATION  
 BIRTH / INDIA LINK / STATUS DATE -> only if sourced

### ASCII MASTER FLOW - PANEL 2/12: Scientific contribution classifier

THEORY -> explanation and prediction  
 OBSERVATION -> recorded phenomenon  
 EXPERIMENT -> controlled test  
 DISCOVERY -> established phenomenon or relation  
 INVENTION / DEVELOPMENT -> device, process or method

### ASCII MASTER FLOW - PANEL 3/12: Nobel category map

PHYSICS | CHEMISTRY | PHYSIOLOGY OR MEDICINE  
 LITERATURE | PEACE -> original framework  
 ECONOMIC SCIENCES -> later memorial prize  
 AWARDING INSTITUTION -> category-specific  
 PEACE -> person or organisation possible

#### ASCII MASTER FLOW - PANEL 4/12: Citation-to-outcome staircase

OFFICIAL CITATION  
DISCOVERY / MECHANISM  
POSSIBLE APPLICATION  
ENGINEERING + COMMERCIALIZATION  
MEASURED POLICY / SOCIAL OUTCOME

#### ASCII MASTER FLOW - PANEL 5/12: Attribution and chronology rail

PRIOR THEORY -> INSTRUMENT -> EXPERIMENT  
PUBLICATION -> REPLICATION -> AWARD  
PATENT / PRODUCT / DEPLOYMENT -> later and separate  
LAUREATE != institution  
AWARD YEAR != discovery year automatically

#### ASCII MASTER FLOW - PANEL 6/12: Physics 2025 quantum-circuit map

SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION  
COLLECTIVE MACROSCOPIC STATE  
BARRIER -> quantum tunnelling  
SPECIFIC ENERGY AMOUNTS -> quantisation  
DEMONSTRATION != fault-tolerant quantum computer

#### ASCII MASTER FLOW - PANEL 7/12: John Clarke clue resolver

BORN -> Cambridge, United Kingdom  
AFFILIATION -> University of California, Berkeley  
CATEGORY / YEAR -> Physics 2025  
SHARE -> one-third with Devoret and Martinis  
MATCH ALL CLUES -> then identify

#### ASCII MASTER FLOW - PANEL 8/12: Chemistry 2025 MOF architecture

METAL IONS -> nodes  
ORGANIC MOLECULES -> linkers  
POROUS CRYSTAL -> cavities  
BUILDING-BLOCK CHANGE -> tunable function  
CAPABILITY / APPLICATION != commercialization

#### ASCII MASTER FLOW - PANEL 9/12: Medicine 2025 tolerance pathway

CENTRAL TOLERANCE -> thymic control layer  
REGULATORY T CELLS -> peripheral control  
Foxp3 -> regulatory-T-cell development  
SELF-TISSUE PROTECTION -> mechanism  
CLINICAL TRIAL != approved therapy

### ASCII MASTER FLOW - PANEL 10/12: MicroRNA regulation rail

DNA --transcription--> mRNA  
microRNA -> complementary mRNA binding  
PROTEIN PRODUCTION -> inhibited  
mRNA DEGRADATION -> possible route  
DISCOVERY != diagnostic or medicine  
MUST REMEMBER: Scientific-history questions separate observation, discovery, theory,...  
MUST REMEMBER: Scientific-history questions separate observation, discovery, theory,...

### ASCII MASTER FLOW - PANEL 11/12: India-linked scientist cards

RAMAN 1930 -> light scattering / Raman effect  
KHORANA 1968 -> genetic code / protein synthesis  
RAMAKRISHNAN 2009 -> ribosome structure and function  
BIRTHPLACE != award-time affiliation  
CITATION != every later application  
CLOSE DISTINCTION: Discovery is not invention, theory is not proof beyond revision, award...  
CLOSE DISTINCTION: Discovery is not invention, theory is not proof beyond revision, award...

### ASCII MASTER FLOW - PANEL 12/12: PYQ and future-status firewall

2026 Q81 -> direct biographical prize route  
BOSE-EINSTEIN / BLUE LED -> Topic 21 direct science owner  
TOPIC 26 -> attribution + citation + maturity support  
2025 OFFICIAL AWARDS -> latest completed science anchors  
4 SEP 2026 -> no invented 2026 laureates  
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use Nobel Foundation or equally...  
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use Nobel Foundation or equally...